

Laboratory Exercise 2.1**EXPLORING MOBILE DEVICE ARCHITECTURES AND OPERATING SYSTEMS**

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In this lab, you will be able to:

1. Discuss the basic hardware architecture of mobile devices and identify key components (CPU, memory, sensors, etc.).
2. Compare the differences in hardware architecture between Android and iOS devices.
3. Explore and analyze the user interfaces and features of both Android and iOS operating systems through emulators.
4. Develop a critical understanding of the UI/UX design principles in mobile operating systems and their impact on user experience

Direction:

1. Architecture Exploration:

- Research and document the architecture of mobile devices (CPU, memory, storage, sensors, etc.).

Mobile Device Architecture

The architecture of mobile devices is a sophisticated integration of hardware components designed to balance high performance with portability and power efficiency. At its core is the System-on-a-Chip (SoC), which integrates the CPU (often utilizing an ARM-based architecture with performance and efficiency cores), the GPU for graphics, and an NPU for AI processing. Memory is typically divided into LPDDR RAM for active processes and NAND Flash storage (UFS or NVMe) for long-term data. Additionally, a diverse array of sensors, including accelerometers, gyroscopes, magnetometers, and biometric scanners (fingerprint/FaceID) allows the device to interact with its physical environment.

Reference: Bithowl. (2025, August 16). Mobile architecture: A comprehensive guide. Medium. <https://medium.com/@bithowl/mobile-architecture-a-comprehensive-guide-1449c7184ee1>

- Compare the hardware architecture of Android and iOS devices. Note key differences in terms of components, performance, and power consumption.

Android vs. iOS

The primary difference between Android and iOS hardware lies in the ecosystem philosophy. Android operates on a fragmented hardware model, running on thousands of different SoCs (Qualcomm, MediaTek, Exynos) and various RAM capacities, which requires the OS to be highly adaptable. In contrast, iOS runs on a closed, standardized architecture using Apple's custom A-series or M-series chips. This tight integration allows iOS devices to achieve superior performance and power efficiency with lower raw RAM specs compared to Android, as the software is specifically tuned for a handful of known hardware configurations.

Reference: Flexiple. (2025). Android vs iOS - A detailed comparison. <https://flexiple.com/compare/android-vs-ios>

- Write a report summarizing the architecture of mobile devices and highlighting the difference between Android and iOS hardware.

In summary, while both platforms share the same fundamental building blocks, their implementation strategies diverge significantly. Android’s architecture is built for diversity and scalability, supporting a wide range of price points and hardware capabilities, which often leads to higher power consumption due to less specialized optimization. iOS’s architecture is built for vertical integration, where Apple controls every transistor and line of code, leading to highly optimized energy cycles and consistent performance across its limited device lineup.

2. Operating System Features:

- Research about Android Studio and Xcode.

Android Studio and Xcode

Android Studio is the official Integrated Development Environment (IDE) for Android, built on JetBrains' IntelliJ IDEA. It supports Kotlin, Java, and C++, offering powerful tools like a flexible Gradle-based build system and a fast emulator. Xcode is Apple's exclusive IDE for iOS, macOS, watchOS, and tvOS development. It uses the Swift and Objective-C languages and is renowned for its Interface Builder and deep integration with the Apple ecosystem, though it is strictly limited to running on macOS hardware.

Reference: SelectHub. (2026, January 5). Android Studio vs Xcode: Which integrated development environments wins in 2026? <https://www.selecthub.com/integrated-development-environment-solutions/android-studio-vs-xcode/>

- Explore the Android OS and iOS features and requirements.

Feature	Android OS	iOS
Kernel Base	Linux (Open-source)	Darwin/Unix (Closed-source)
Hardware Compatibility	Massive variety of manufacturers (Samsung, Google, etc.)	Exclusive to Apple hardware (iPhone/iPad)
Minimum RAM Needs	Higher (typically 4GB-16GB) due to Java-based runtime	Lower (typically 4GB-8GB) due to native optimization
App Sourcing	Google Play Store, third-party stores, and APK sideloading	Apple App Store (Limited sideloading in specific regions)

- Analyze the user interface (UI) and key features (e.g., notifications, multitasking, file management).

Both operating systems offer advanced multitasking, but they handle it differently: Android allows true background processing and split-screen on most devices, while iOS uses a stricter suspended state to save battery. Notifications on Android are generally more granular and bundled by app, whereas iOS focuses on a chronological "Notification Center" and "Live

Activities." File management is historically more open on Android, allowing direct folder access, while iOS uses a sandboxed "Files" app that prioritizes security over direct system access.

- Compare the user interface and user experience (UI/UX) design principles between Android and iOS. Document the differences and similarities.

Aspect	Android	iOS	Comparison
Visual Philosophy	Tactile Surface: Uses "Digital Paper" where elements have physical weight, shadows, and Z-axis depth.	Clarity & Deference: Prioritizes content over the interface. Uses translucency (blur) and flat elements.	Difference: Android feels more "physical"; iOS feels more like "glass" or layers of light.
Navigation	Universal Back: Relies on a system-wide back gesture or button that works across all apps.	Contextual Back: Uses a specific "Back" button in the top-left corner or a right-swipe from the edge.	Difference: Android has a consistent global exit; iOS navigation is app-dependent.
Main Menu	Navigation Drawer: Often uses the "Hamburger" menu (top-left) to hide secondary navigation.	Tab Bar: Prefers a persistent bottom bar for the most important app sections.	Difference: iOS prioritizes immediate access; Android prioritizes screen real estate.
Action Buttons	Floating Action Button (FAB): A promoted circular button for the primary action (e.g., "Compose").	Top-Right Action: The primary action (e.g., "Done" or "Edit") is usually text-based at the top.	Difference: Android highlights the primary action via shape; iOS via placement.
Typography	Roboto / Noto: Optimized for readability across a massive variety of screen sizes.	San Francisco: Designed for high legibility and adaptability to varying font weights.	Similarity: Both use proprietary sans-serif fonts optimized for mobile screens.

Touch Targets	Standard minimum is generally 48x48 dp .	Standard minimum is generally 44x44 pt .	Similarity: Both emphasize large enough targets to prevent "fat-finger" errors.
Iconography	Prefers bold, filled icons with distinct geometric shapes.	Prefers thin, outlined icons that feel light and airy.	Difference: Android icons feel more sturdy; iOS icons feel more elegant.

3. Task:

- Write a comparison of Android and iOS in terms of architecture, UI/UX, and key features. Include screenshots from the emulator/simulator where necessary.

Feature	Android	iOS
Kernel / Architecture	Modified Linux Kernel; Open-source.	Darwin (Unix-based) Kernel; Closed-source.
Hardware Range	Wide variety (Samsung, Google, Xiaomi, etc.).	Exclusive to Apple (iPhone/iPad).
Primary Languages	Kotlin, Java.	Swift, Objective-C.
IDE	Android Studio (Windows, Mac, Linux).	Xcode (macOS only).
UI Design Language	Material Design (Tactile, bold, depth).	Human Interface Guidelines (Minimalist, flat).
Navigation	System back button/gesture; Navigation drawer.	Top-left back button; Bottom tab bars; Edge-swipe.
Customization	High (Launchers, widgets, system-level tweaks).	Low to Moderate (Focus on consistency and simplicity).
File Management	Direct access to file system/internal storage.	Sandboxed; Limited access via Files app.
Performance/Power	Varies by hardware; more background freedom.	Highly optimized; strict background limitations.

The fundamental difference between Android and iOS lies in the balance between open flexibility and controlled optimization. Android's architecture, rooted in the open-source Linux kernel, is designed for hardware diversity and user agency, offering an open file system, extensive customization through Material Design, and true multitasking capabilities across various manufacturers. Conversely, iOS utilizes a closed, vertically integrated architecture where Apple's proprietary hardware and Darwin-based software are specifically tuned for one another, resulting in superior power efficiency and a highly predictable UI/UX guided by minimalist Human Interface Guidelines. Ultimately, while Android provides a "pocket computer" experience characterized by freedom and scale, iOS offers a streamlined, secure, and cohesive environment where performance is maximized through strict software-hardware synergy.

Expected Deliverables:

- A report on mobile device architecture.
- A comparison document between Android and iOS OS and their features.